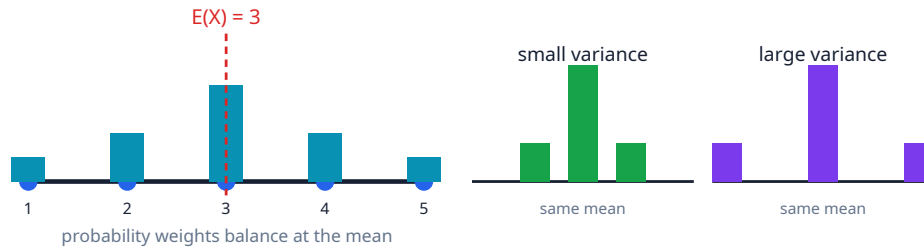


# 04 EXPECTATION AND VARIANCE (notes p086-p111)

## 1. The three parameters (p086-p087)

Mean is the balance point; variance is squared spread



### Expectation

$E[h(X)] = \text{sum or integral of } h(x) \text{ times weight}$

### Variance

$\text{Var}(X) = E(X^2) - [E(X)]^2$

shifts preserve variance; multiplying by a scales variance by  $a^2$

Mean as balance point and variance as spread

|                     |                               |
|---------------------|-------------------------------|
| MEAN $\mu$          | centre / location             |
| VARIANCE $\sigma^2$ | spread                        |
| SD $\sigma$         | spread, in the ORIGINAL units |

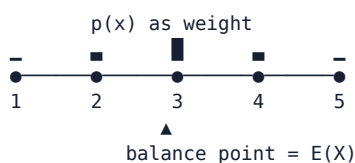
"knowledge of the numerical values of these parameters gives the researcher quick insight into the nature of the variable"

## 2. Expectation of a discrete rv (p088)

$$E(X) = \mu_X = \sum_{x \in D} x \cdot p(x)$$

"the weighted average of the values, weighted by their probabilities"

the picture: a balance beam. values on the x-axis, probabilities as weights. the mean is the BALANCE POINT.



## 3. Worked: a fair die (p089)

$$p(x) = 1/6, \quad x = 1..6$$

$$E(X) = 1(1/6) + 2(1/6) + 3(1/6) + 4(1/6) + 5(1/6) + 6(1/6)$$

$$= (1+2+3+4+5+6)/6 = 21/6 = 7/2 = 3.5$$

3.5 is not a face on the die. The mean need not be an achievable value. It is the long-run average.

## 4. Expectation of a function of X (p090)

$$E[h(X)] = \sum_{\{x\}} h(x) p(x) \quad (\text{discrete})$$

$$E[h(X)] = \int h(x) f(x) dx \quad (\text{continuous})$$

$$E[h(X)] \neq h(E(X)) \quad \text{in general}$$

you must apply  $h$  to EACH value, then weight, then sum.  
you may NOT plug the mean into  $h$ .

## 5. Properties of the mean (p092-p093)

THEOREM  $E[c] = c$        $E[cX] = c E[X]$

proof of  $E[c]$ :

$$E[c] = \sum c f(x) = c \sum f(x) = c(1) = c \quad \checkmark$$

proof of  $E[cX]$ :

$$E[cX] = \sum c x f(x) = c \sum x f(x) = c E[X] \quad \checkmark$$

EXERCISE (p093): prove  $E[aX + b] = a E[X] + b$

$$\begin{aligned} E[aX+b] &= \sum (ax+b)f(x) = a \sum x f(x) + b \sum f(x) \\ &= a E[X] + b(1) = a E[X] + b \quad \checkmark \end{aligned}$$

$E$  is LINEAR, and it is linear with NO independence needed:  
 $E[X+Y] = E[X] + E[Y]$  ALWAYS  
 $E[aX+b] = aE[X]+b$  ALWAYS

## 6. Worked: the p094 triple (p094) - re-asked on ETE papers

$$\begin{array}{lcl} x & : & -3 \quad 6 \quad 9 \\ P(X=x) & : & 1/6 \quad 1/2 \quad 1/3 \end{array}$$

$$\begin{aligned} E(X) &= (-3)(1/6) + 6(1/2) + 9(1/3) \\ &= -0.5 + 3 + 3 = 11/2 = 5.5 \end{aligned}$$

$$\begin{aligned} E(X^2) &= 9(1/6) + 36(1/2) + 81(1/3) \\ &= 1.5 + 18 + 27 = 93/2 = 46.5 \end{aligned}$$

$$\begin{aligned} E((2X+1)^2) : \text{expand first} &= E[4X^2 + 4X + 1] \\ &= 4 E(X^2) + 4 E(X) + 1 \\ &= 4(46.5) + 4(5.5) + 1 \\ &= 186 + 22 + 1 = 209 \end{aligned}$$

THE METHOD: EXPAND, then use linearity term by term.  
Do NOT compute the 3 values of  $(2x+1)^2$  by hand first (you can, but the algebra route is faster and safer).

sanity:  $(2x+1)^2$  for  $x=-3,6,9 = 25,169,361$   
 weighted:  $25/6 + 169/2 + 361/3 = 4.17+84.5+120.3 = 209 \quad \checkmark$

## 7. Variance (p095-p100)

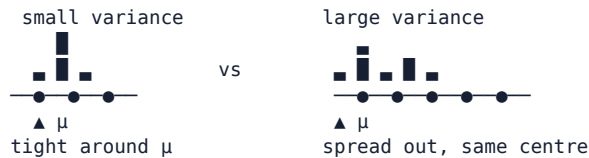
$$\begin{aligned} \text{Var}(X) &= \sigma^2 = E[(X - \mu)^2] \\ \text{SD} &= \sigma = \sqrt{\text{Var}(X)} \end{aligned}$$

$\text{Var}(X)$  is ALWAYS non-negative (it is an expectation of a SQUARE).

the idea: the SQUARED deviation from the mean, weighted by probability.  
 large  $\sigma^2$   $\Leftrightarrow$  more squared spread around  $\mu$   
 small  $\sigma^2$   $\Leftrightarrow$  less squared spread around  $\mu$

Variance measures spread. It is not a universal equivalence to predictability, and it is unrelated to the estimator property called consistency.

TWO DISTRIBUTIONS, SAME MEAN, DIFFERENT VARIANCE:



## 8. The shortcut formula (p101-p103) - and its proof

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

PROOF:

$$\begin{aligned} \text{Var}(X) &= E[(X-\mu)^2] \\ &= E[X^2 - 2\mu X + \mu^2] && \text{expand the square} \\ &= E[X^2] - 2\mu E[X] + \mu^2 && \text{linearity} \\ &= E[X^2] - 2\mu\mu + \mu^2 && \text{since } E[X] = \mu \\ &= E[X^2] - 2\mu^2 + \mu^2 \\ &= E[X^2] - \mu^2 && \checkmark \end{aligned}$$

THE TWO-STEP ROUTINE, ALWAYS:

1.  $E(X)$
2.  $E(X^2)$
3.  $\text{Var} = E(X^2) - [E(X)]^2$

You almost never use the definition directly.

## 9. Variance transforms (p104-p106)

THEOREM  $\text{Var}[c] = 0$        $\text{Var}[cX] = c^2 \text{Var}[X]$   
 PROPOSITION  $\text{Var}[aX+b] = a^2 \sigma^2$        $\sigma_{\{aX+b\}} = |a| \sigma_x$

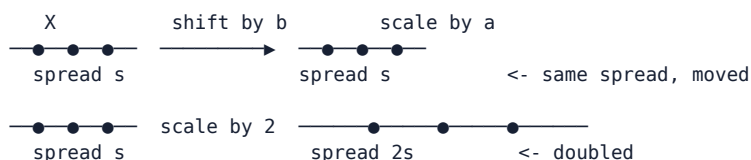
special cases:

$$\begin{aligned} \sigma_{\{aX\}} &= |a| \sigma_x && \text{scaling changes spread by } |a| \\ \sigma_{\{X+b\}} &= \sigma_x && \text{SHIFTING changes nothing} \end{aligned}$$

WHY  $|a|$  AND NOT  $a$

$a$  may be negative, but a standard deviation cannot be.  
 The sign of  $a$  does not affect spread, only scale.

the picture:



## 10. Worked: freezer dealer (p107-p108) - the 6-part chain

|        |        |      |      |              |
|--------|--------|------|------|--------------|
| $x$    | : 13.5 | 15.9 | 19.1 | (cubic feet) |
| $p(x)$ | : .2   | .5   | .3   |              |

$$\begin{aligned} \text{(a) } E(X) &= 13.5(.2) + 15.9(.5) + 19.1(.3) \\ &= 2.70 + 7.95 + 5.73 = 16.38 \end{aligned}$$

$$\begin{aligned} E(X^2) &= 182.25(.2) + 252.81(.5) + 364.81(.3) \\ &= 36.45 + 126.405 + 109.443 = 272.298 \end{aligned}$$

$$V(X) = 272.298 - (16.38)^2 = 272.298 - 268.3044 = 3.9936$$

- (b) price =  $25X - 8.5$   
 $E(25X - 8.5) = 25(16.38) - 8.5 = 409.5 - 8.5 = 401$
- (c)  $V(25X - 8.5) = (25)^2 V(X) = 625 \times 3.9936 = 2496$   
 (the  $-8.5$  vanishes from the variance)
- (d) actual capacity  $h(X) = X - .01X^2$   
 $E[h(X)] = E(X) - .01 E(X^2) = 16.38 - .01(272.298)$   
 $= 16.38 - 2.723 = 13.657 \approx 13.66$

PART (d) IS THE TRAP PART.  
 $E[X - .01X^2]$  is NOT  $16.38 - .01(16.38)^2$ .  
 You must use  $E(X^2)$ , NOT  $[E(X)]^2$ .  
 This is  $E[h(X)] \neq h(E(X))$  in numerical form.

## 11. Worked: magazine orders (p109-p111) - the decision question

x : 1 2 3 4 5 6  
 p(x) : 1/15 2/15 3/15 4/15 3/15 2/15

store pays \$1 per copy, sells at \$2, leftovers worthless.  
 is it better to order 3 or 4?

net revenue as a function of demand x:

x : 1 2 3 4 5 6  
 h3(x) : -1 1 3 3 3 3 <- order 3: plateaus at 3  
 h4(x) : -2 0 2 4 4 4 <- order 4: plateaus at 4  
 p(x) : 1/15 2/15 3/15 4/15 3/15 2/15

WHY h3 plateaus: order 3, cost 3, revenue 2 per copy SOLD (max 3 copies)

x=1: sold 1, revenue 2, cost 3 -> -1

x=2: sold 2, revenue 4, cost 3 -> +1

x=3: sold 3, revenue 6, cost 3 -> +3

x=4: can only sell 3, revenue 6, cost 3 -> +3 (plateau: demand > stock)

$$E[h3(X)] = (-1)(1/15) + (1)(2/15) + (3)(3/15) + (3)(4/15) + (3)(3/15) + (3)(2/15)$$

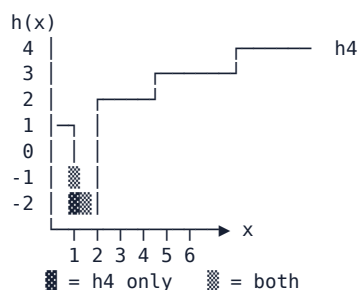
$$= (-1 + 2 + 9 + 12 + 9 + 6)/15 = 37/15 = 2.4667$$

$$E[h4(X)] = (-2)(1/15) + (0)(2/15) + (2)(3/15) + (4)(4/15) + (4)(3/15) + (4)(2/15)$$

$$= (-2 + 0 + 6 + 16 + 12 + 8)/15 = 40/15 = 2.6667$$

ORDER 4: higher expected revenue (2.67 vs 2.47)  
 This is a DECISION question: express profit as h(X),  
 take E of each option, compare.

the two profit curves:



Expectation and variance done. Next: continuous random variables (p112-p147).